

Introduction To Web Technologies

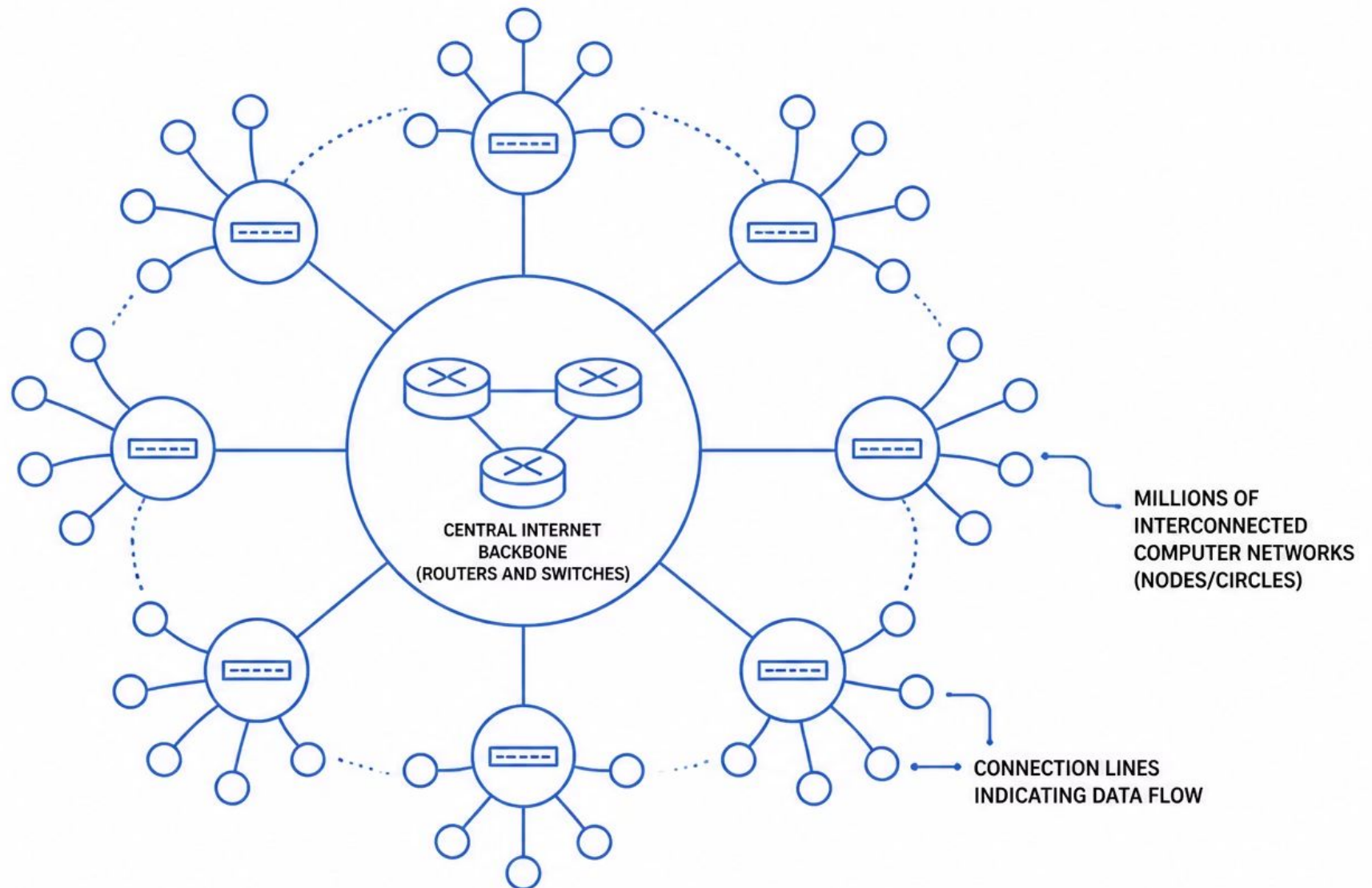
Client Side Technologies

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What is Internet?

Millions of computer networks connected to each other via the 'Internet Backbone'



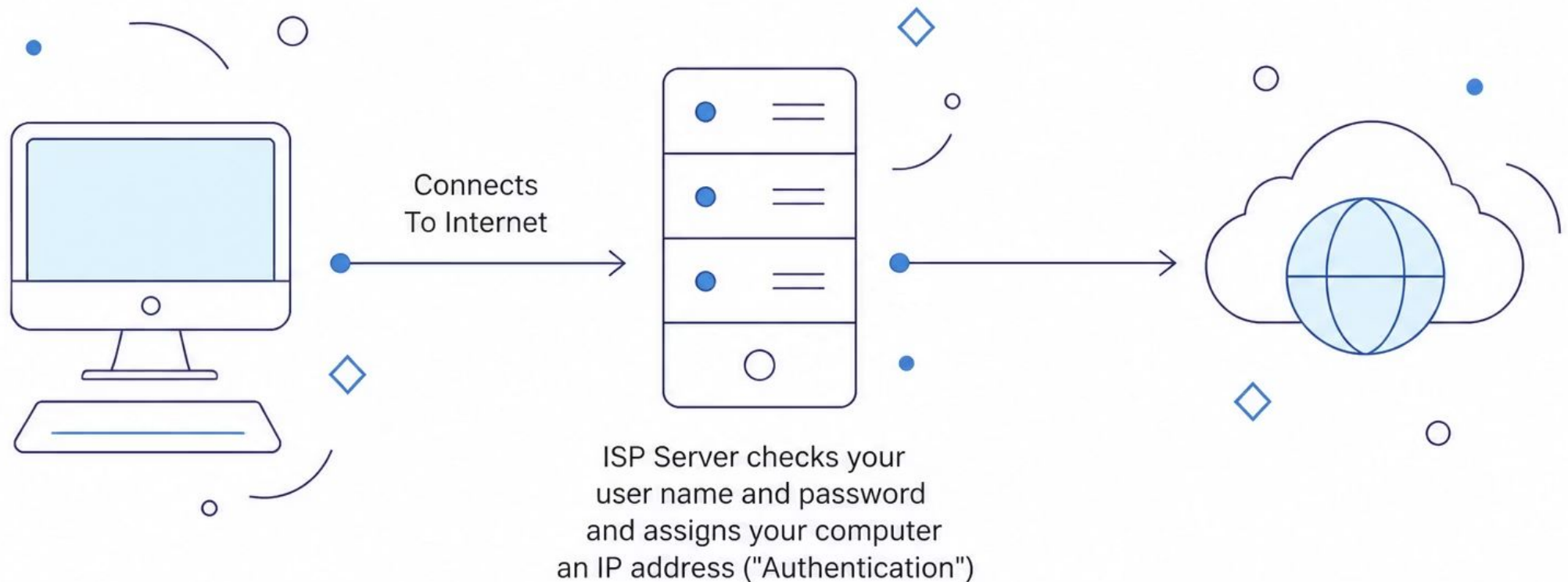
What is Internet? (Cont.)

The Internet Backbone:

- The Internet Backbone is an infrastructure of fiber optics cable owned by private companies.

How does your computer connect to the Internet?

Your computer Connects To Internet



Internet Service Provider

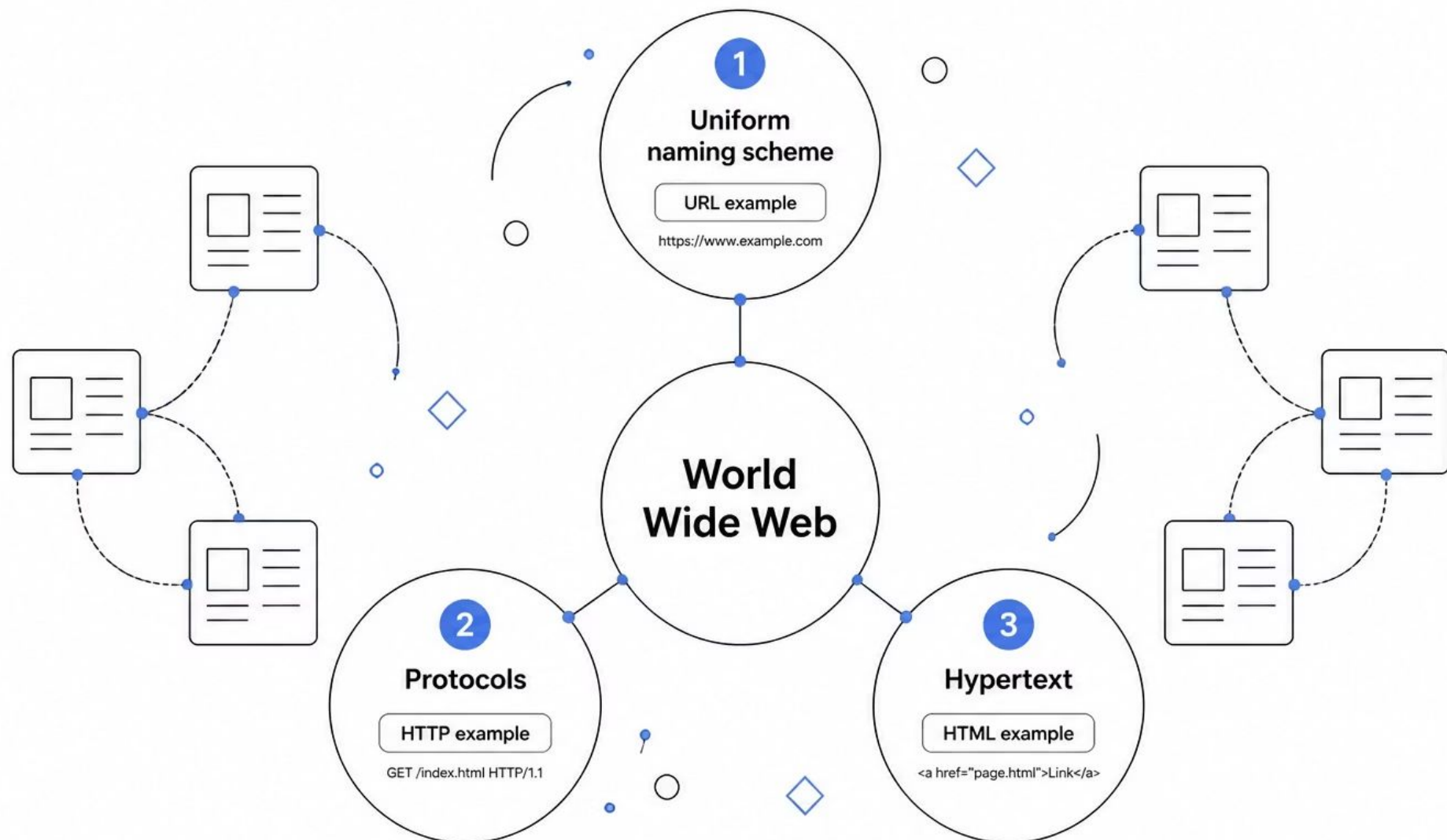
What is the World Wide Web?

The World Wide Web (Web):

- Is a network of information resources (or collection of websites).

The Web relies on three mechanisms to make these resources readily available to the widest possible audience:

- A uniform naming scheme for locating resources on the Web (e.g., URL).
- Protocols: for access to named resources over the Web (e.g., HTTP).
- Hypertext: for easy navigation among resources (e.g., HTML).



What is the World Wide Web? (Cont.)

Hypertext:

- Is text displayed on a computer with references (Hyperlinks) to other text that the reader can immediately access, usually by a mouse click. Hypertext may contain tables, images, videos and other resources. The most extensive example of hypertext today is the World Wide Web.

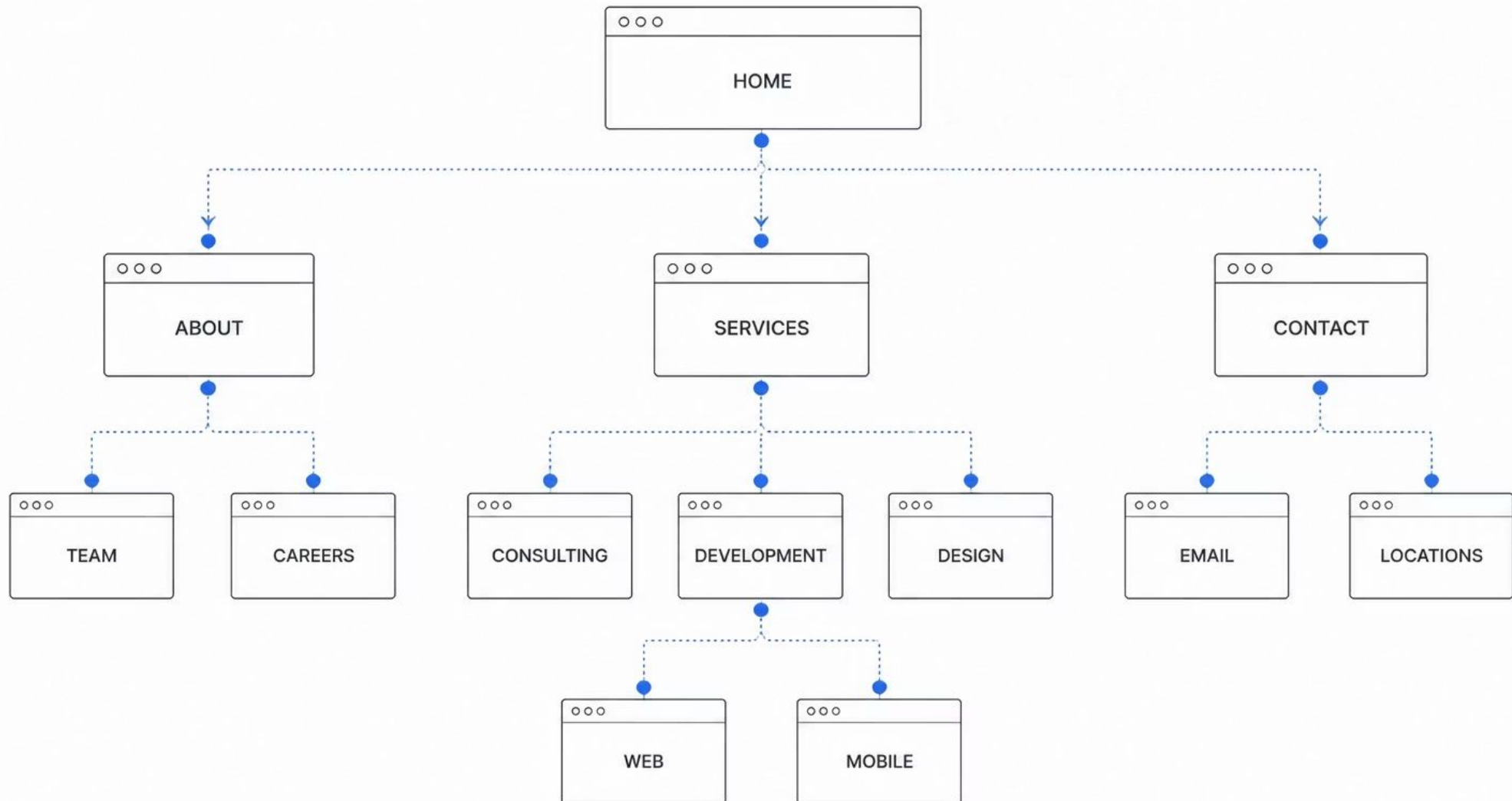
Hyperlink (or link):

- Is a reference to a document or a resource on the web.

What is Website?

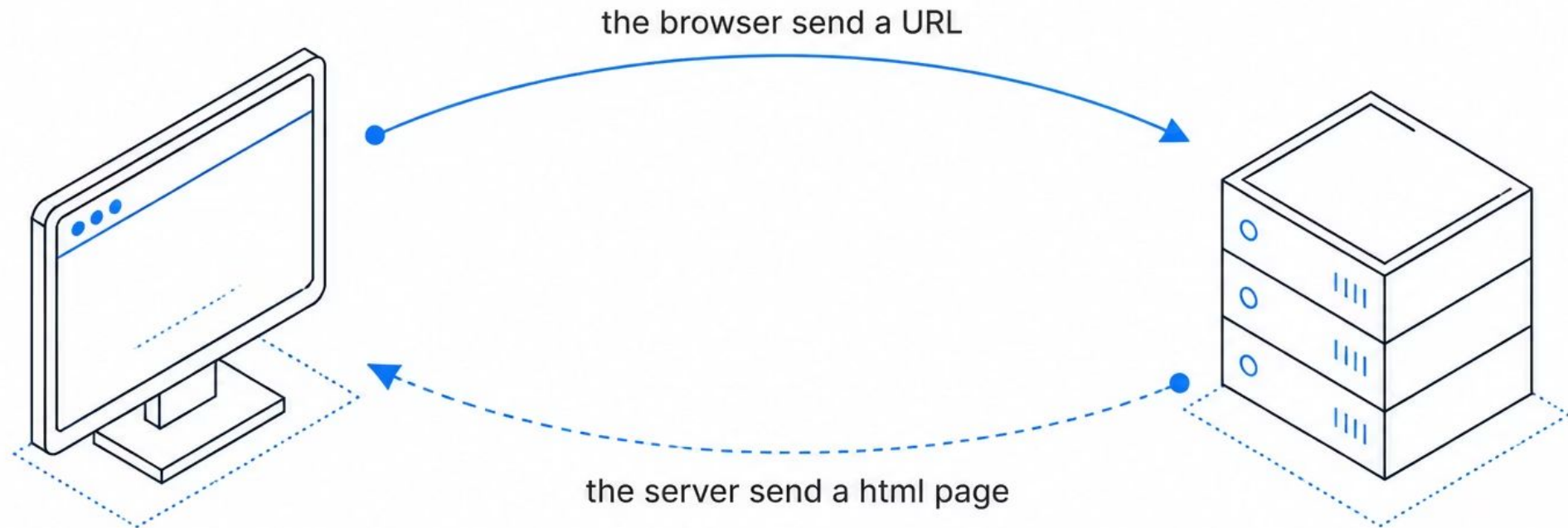
- A website is a collection of related and linked web pages via hyperlinks, images, videos or other digital assets that are addressed with a common domain name or IP address.
- A web page is a document, typically written in Hypertext Markup Language (e.g. HTML).
- All publicly accessible websites collectively constitute the World Wide Web.
- The pages of a website can usually be accessed from (Home Page) which is web page has a simple Uniform Resource Locator (URL).

What is Website? (Cont.)



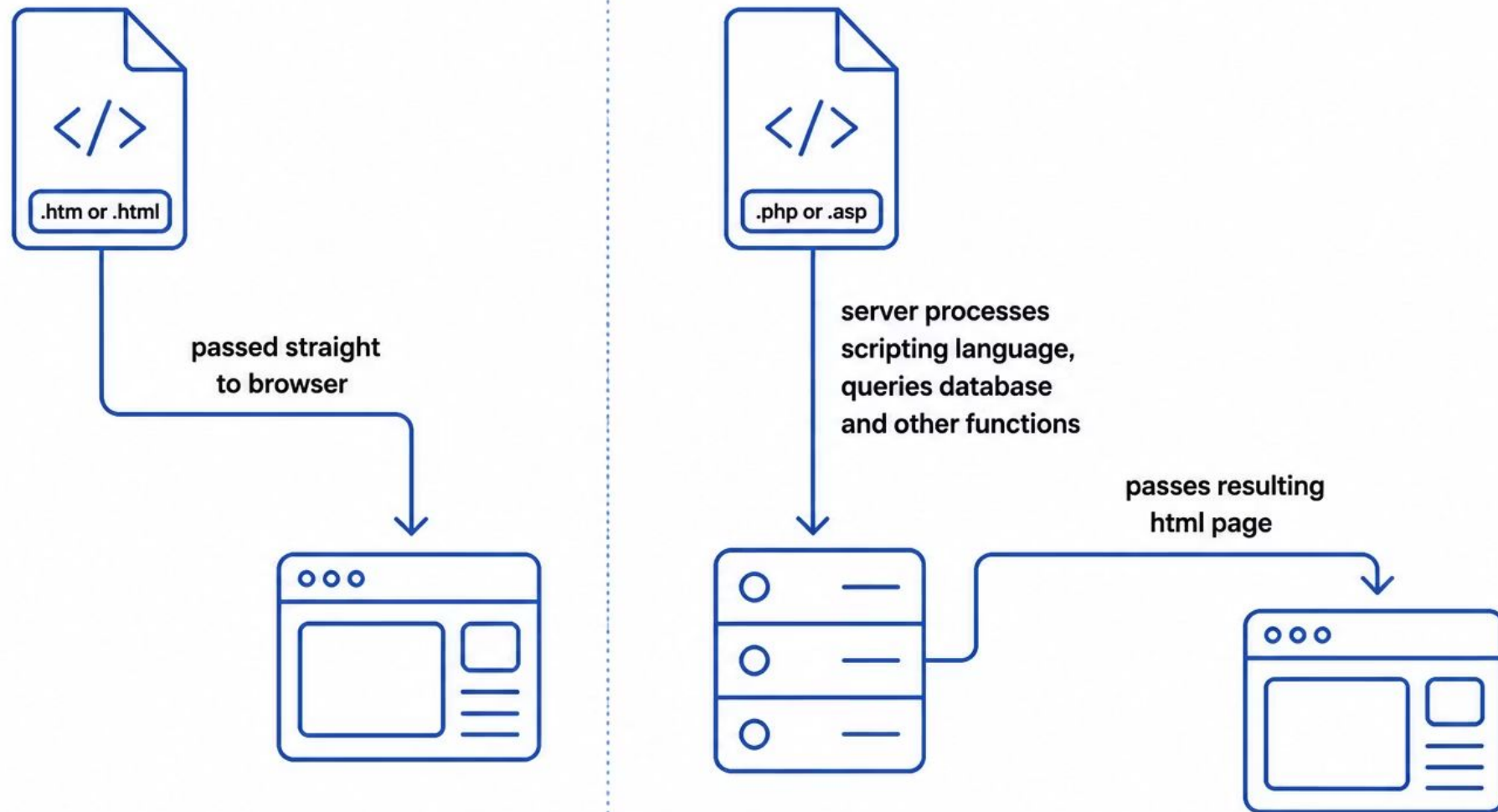
What is Website? (Cont.)

Client / Server architecture:



What is Website? (Cont.)

Static and dynamic web sites:

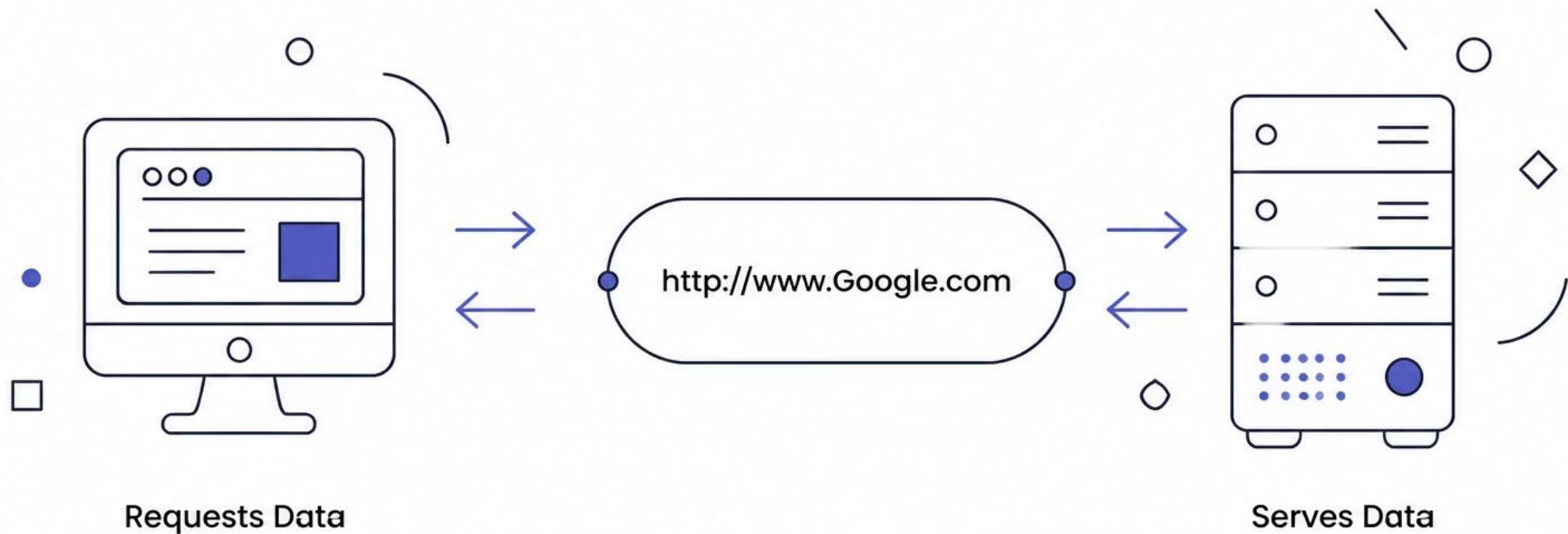


What is URL?

URL (Uniform Resource Locator):

- Unique address of a document or a resource on the internet.
- You type a website address or "URL" into your browser.

Client / Server

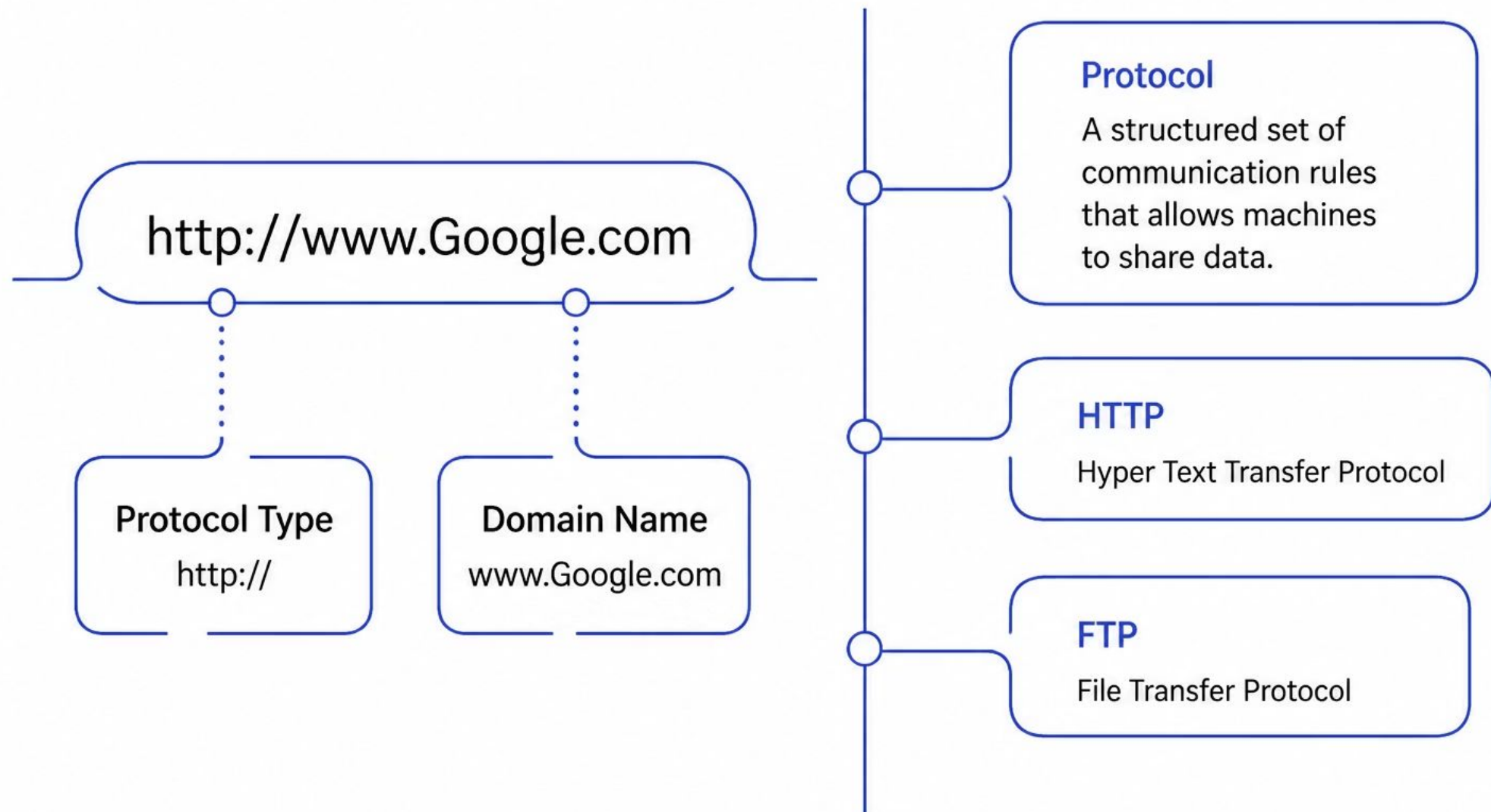


What is URL? (Cont.)

URL Have two parts:

- HTTP = Hyper Text Transfer Protocol
- FTP = File Transfer Protocol

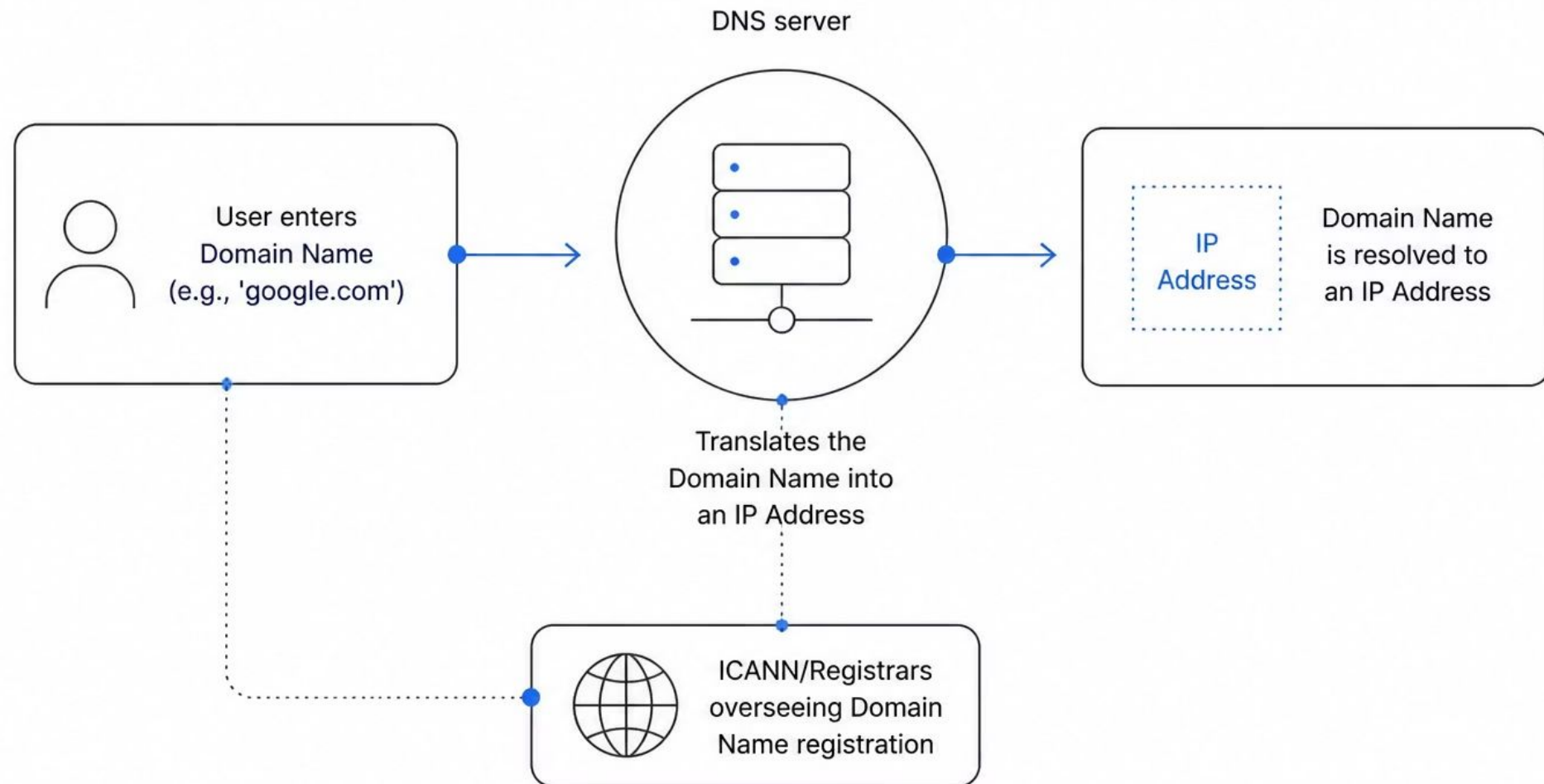
A "Protocol" is a structured set of communication rules that allows machines to share data.



What is URL? (Cont.)

Domain Name:

- The name that identifies an Web site. For example: "google.com" is the domain name of Google's Web site.
- To register a domain name internationally, you must buy it from ICANN or ICANN-Accredited Registrars.
- Because the Internet is based on IP addresses, not domain names, every Web server requires a Domain Name System (DNS) server to translate domain names into IP addresses.



Addressing Schemes

1

Server Addresses

Uniquely identifies a specific computer. Ex.:131.107.1.240

2

Website Domain Name

Uniquely refers to a specific server address. Ex.: iti.net.eg

3

Website URL

Protocol + Domain name. Ex.: http:// www.iti.net.eg

4

Resource Addresses (Resource URL)

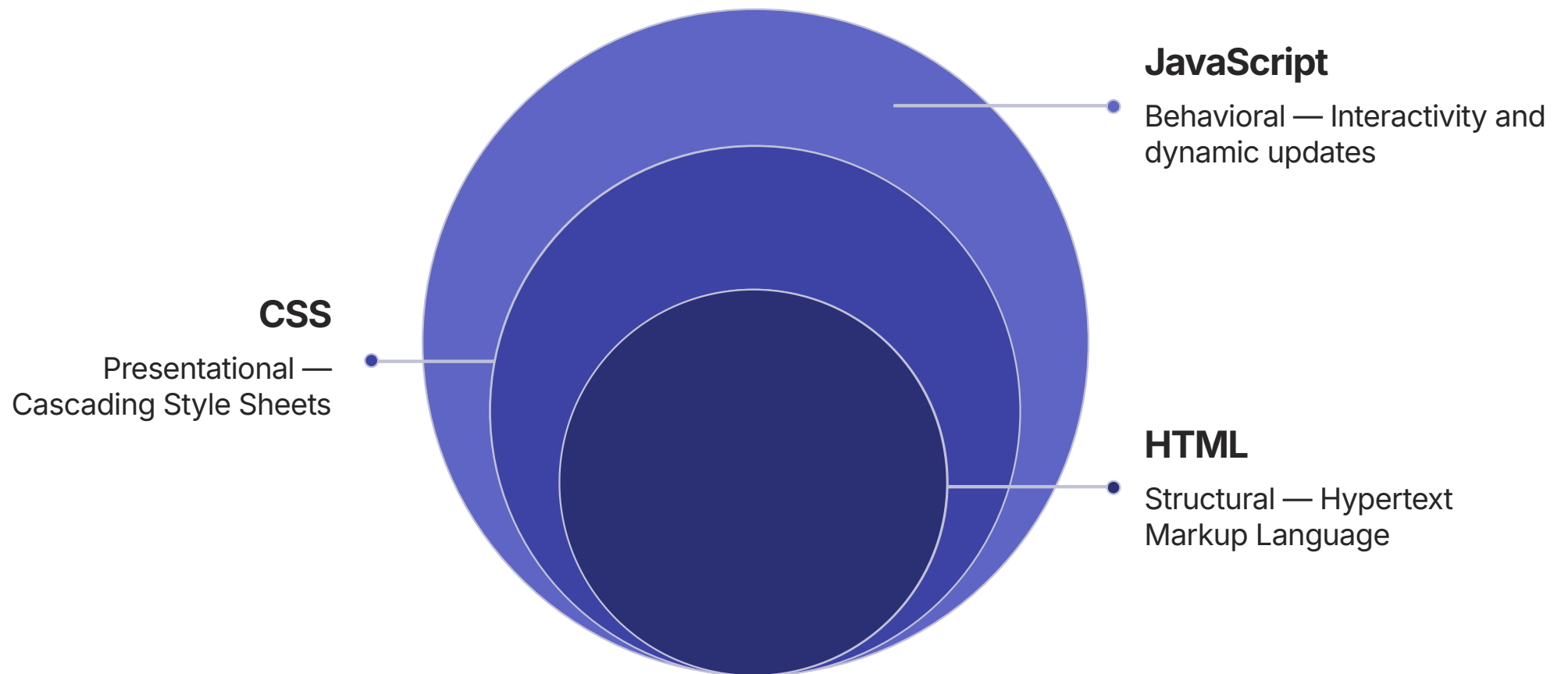
Example : http://www.iti.gov.eg/itiprofile/assuit.jpg

5

E-Mail Addresses

Must be unique. Example: hsaad@iti.net.eg

Technologies used in Web and covered by this course



HTML - Structural

- Hypertext Markup Language
- Structure of Page

CSS - Presentational

- Cascading Style Sheets
- Presentation/Styling

JavaScript - Behavioral

- Interactivity with User
- Dynamic Updates in a Web Page

Self Study..



What is ICANN?



What is IANA?



What is W3C?

Thank You...

Questions ?

